

A blue triangle graphic pointing to the right, positioned to the left of the text.

P320HVN02.3
No display

MQABB0

Topic

- **Characterize**
- **Contain**
- **Cause**
- **Corrective action**
- **Closure**

Characterize

■ General Information

Detailed information for this issue

End Customer	JUNGLIM	SI Maker	
Panel Model Name	P320HVN03.2	Defect Location	L/R
Defect Symptom	2026/01/05	Serial Number	3B5F5FB7SGZZZS1300 (25/34) 3B5F6FF3KRZZZS1300 (25/33)

■ Description

After turning on the defect panels, we found the symptom was ND as customer complaint.



Contain

- Root cause : 3B5F5FB7SGZZZS1300 (25/34)
 - Using IR to measure C/B temp. the U701 high temp. was found.
 - Measuring the ST/LC/HC output waveform were abnormal.
 - As above data, the no display issue should be relate to level shift.

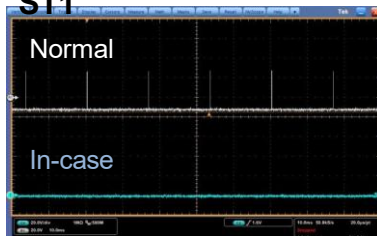
In-case



Normal



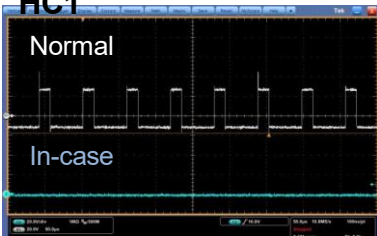
ST1



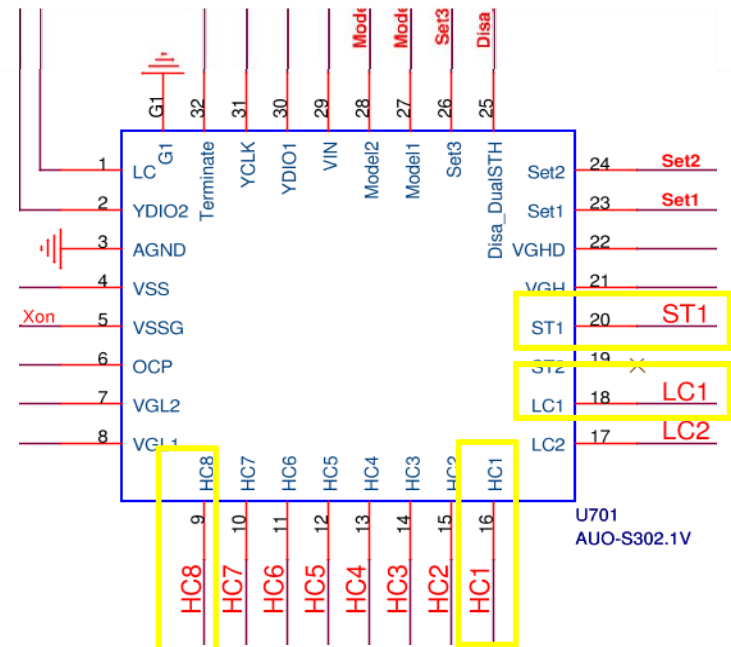
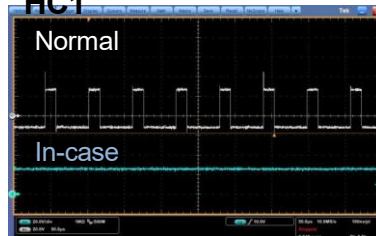
LC1



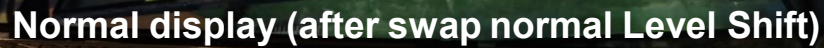
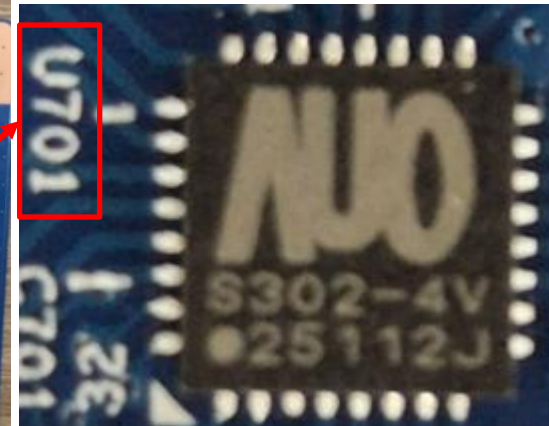
HC1



HC1



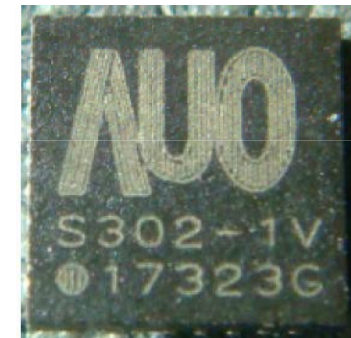
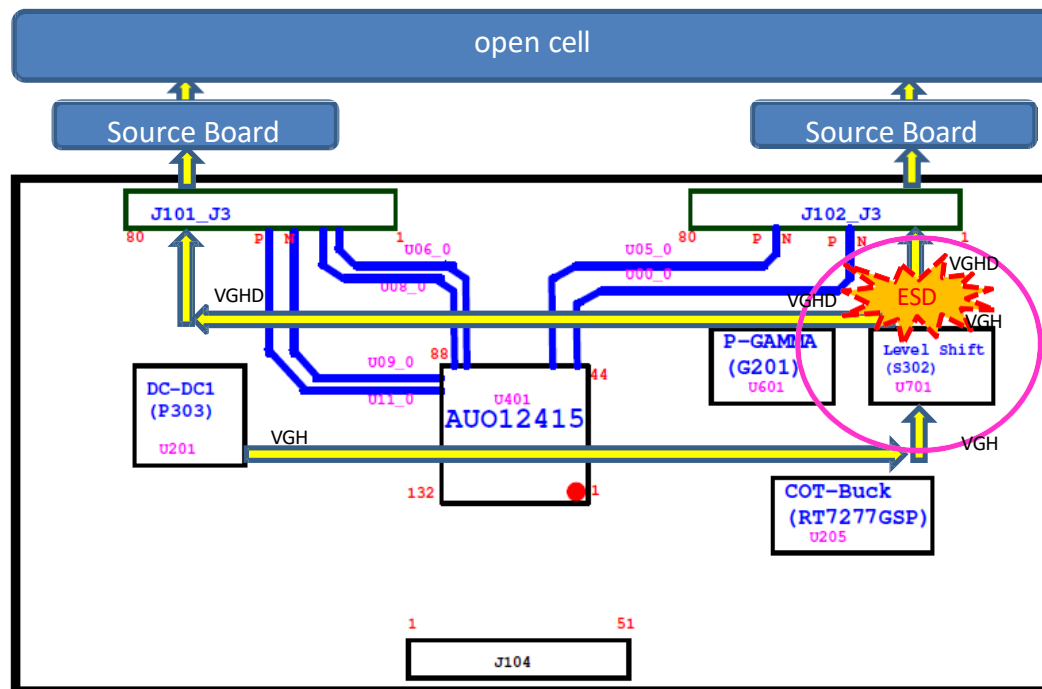
After swap normal Level Shift, the display is normal.



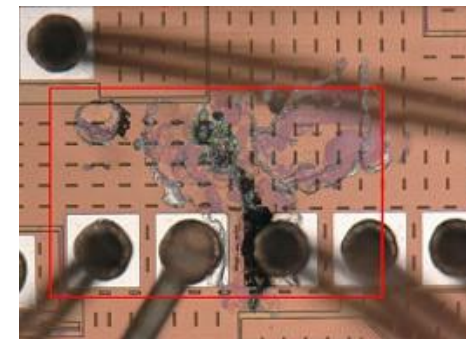
Cause

■ Root Cause

- The sample was performed the de-cap inspection and the result was IC Melt.
- Check the defect was cause by ESD Damage.



IC Melt VGH Circuit



Cause

■ Status

- Check level shift IC ESD Spec : ± 8 KV
- Vender Test Report 8KV / 400V Pass



AUO S302.1V

General Description

The S302.1V has a highly integrated 14 channel output for using in Gate In Panel (GIP) or Gate On Panel (GOP), thin-film transistor (TFT), liquid-crystal displays (LCDs)

The S302.1V provide a condense logic that generates 14 output signals from the several input signals provided by the timing controller. With dedicated design, it prevents from any abnormal panel operation, no matters output short to GND or signals failure.

These outputs are swings from 45V to -20V and high slew rate and high current driving ability which is fitted for any kind of GOP/GIP/GOA panel. Low Rds,on design can achieve thermal reduction. Moreover, thermal protection is also designed in this device.

The S302.1V is available in a 32-pin thin QFN package with a maximum thickness of 0.8mm for ultra-thin LCD panels

Features

- ◆ 2.6V to 5.5V Input Supply Range
- ◆ 14-Channel Condense Level Shifter
45V for High Level
-20V for Low Level
- ◆ ESD Gun ± 8 KV Level on Every Pin
- ◆ Thermal Temperature Lower
- ◆ Suitable for 4/6/8 Phase Applications
- ◆ 1 or 2 Line and Pre-charge Function Provided
- ◆ Over Temperature Protection
- ◆ Short Circuit Protection
- ◆ 32-pin QFN(4mmx4mm) Package
- ◆ RoHS Compliant and 100% Lead (Pb)-Free

1-3. Test Conditions

Test Conditions	Sample Size	Model
RR=1.5K Ω , C=100pF	3	HUMAN BODY MODEL (HBM))

1-4. Assured Capability

ESD Sensitivity	Pin Combination
Pass 8000V	ALL-VSS, ALL-VCC, VCC-VSS

2-3. Test Conditions

Test Conditions	Sample Size	Model
RR=0 Ω , C=200pF	3	MACHINE MODEL (MM)

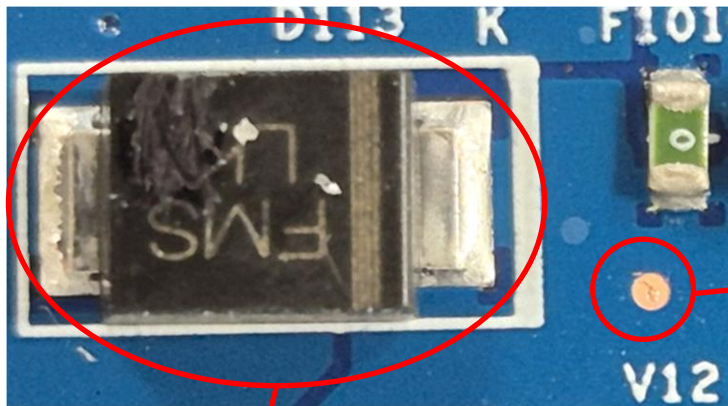
2-4. Test Results

ESD Sensitivity	Pin Combination
Pass 400V	ALL-VSS, ALL-VCC, VCC-VSS

Contain

■ Root cause : 3B5F6FF3KRZZZS1300 (25/33)

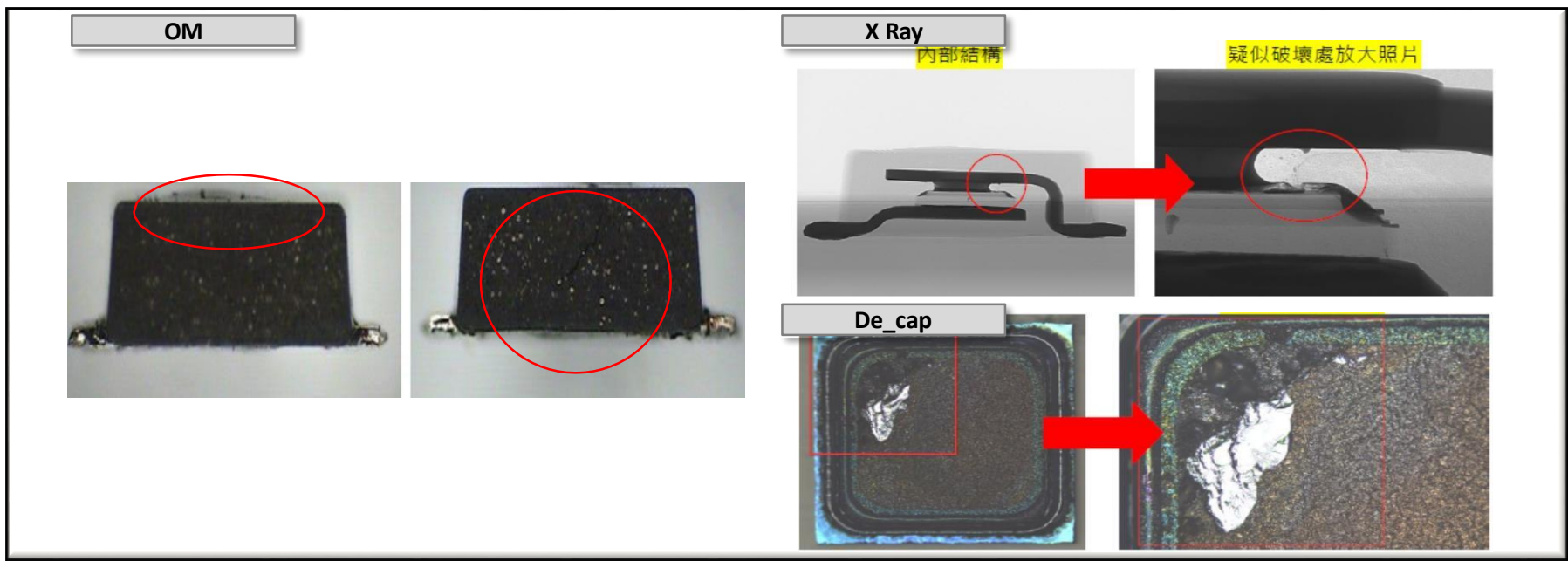
- Turn on the panel, the no display failure was duplicated as customer complained.
- To measure V12, it was found short to GND.
- Check D113 function fail.



Cause

■ Vender analysis

- OM check : deformation was found. Suspect it was caused by EOS/ ESD damage.
- X Ray Check and De-cap : ESD/EOS damage was confirmed.



■ Summary

In order to prevent the situation of ESD/EOS, should prevent the following condition.

1. Hot plug: It should avoid connector plug in/out when power on condition.
2. Poor GND connect: It is easy caused ESD when the GND connect of tester is poor.
3. Unstable power source: It is easy generator noise pulse to cause ESD/EOS when Power source is turn on.
4. Ionizer fan function abnormal: Easy cause ESD/EOS when Ion fan is abnormal.

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Bright Innovation · Amazing Life

The logo for AUO, featuring the letters 'AUO' in a bold, white, sans-serif font. The 'A' and 'U' are connected, and the 'O' is a simple circle. The logo is centered on a dark blue background that has a diagonal split, with a lighter blue section on the left and a darker blue section on the right. The background also features a subtle grid pattern in the upper right and lower right corners.

AUO

Innovating Life